

Research Article

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Estimating positional plus-minus in the NBA

<https://doi.org/10.1515/jqas-2022-0120>

Received December 25, 2022; accepted April 3, 2024;

published online April 25, 2024

Abstract: Plus-minus is a widely used performance metric in sports. Players with high plus-minus ratings are often considered more efficient than others. While numerous plus-minus models have emerged since the introduction of adjusted plus-minus in 2004, most of these metrics focus on evaluating player performance at the individual level. In the present study, we follow the plus-minus framework and adopt a hierarchical Bayesian linear model to estimate plus-minus at the position level in the NBA from 2015–16 to 2021–22. Results show that players with versatile offensive skills and big players who defend the paint area are the most valuable offensive and defensive contributors respectively. We also find that the gaps in offensive plus-minus between offensive position groups have decreased over time. Overall, our analysis offers valuable information regarding average positional values in the NBA, allowing more objective player comparisons within position groups. We also show improved prediction accuracy in player plus-minus when factoring in player positions.

Keywords: basketball; Bayesian analysis; player positions; player evaluation

1 Introduction

How to objectively evaluate player performance is a core question in the field of sports analytics. So far, scholars have proposed a variety of metrics to study player and team performance (Baghal 2012; Franks et al. 2015). Among these, the plus-minus rating is perhaps the most popular one (Engelmann 2011; Fellingham 2022; Hvattum 2019; Macdonald 2012; Rosenbaum 2004; Sill 2010; Winston 2012). In general, plus-minus intends to measure how players contribute to team offense and defense while they are on the

court (Grasseti et al. 2021; Sabin 2021). If players are effective in helping teams win, their plus-minus should be high. Due to the nature of sport competition, the plus-minus metric is often divided into two parts, offensive plus-minus and defensive plus-minus. Offensive plus-minus measures players' contributions to team offense, while defensive plus-minus quantifies players' contributions to team defense (Macdonald 2011).

Prior studies of plus-minus predominately focus on estimating plus-minus for individual players (Deshpande and Jensen 2016). For example, Plus-minus metrics such as regularized adjusted plus-minus (RAPM) (Engelmann 2011; Sill 2010), adjusted plus-minus (Rosenbaum 2004), ESPN's real plus-minus (Ilardi and Engelmann 2014), and estimated plus-minus (Snarr 2022), have developed over the past decade to measure player efficiency. In this study, we explore NBA player plus-minus ratings at the position level by introducing a hierarchical structure to previous plus-minus models (Sabin 2021; Thomas et al. 2013; Yurko et al. 2019).

Understanding positional values is pivotal to NBA teams in trades and free agency as these values can serve as average group ratings in player evaluation (Whitehead 2019). For example, NBA teams may compare player plus-minus to positional plus-minus in order to more accurately assess player efficiency. Studying positional values over several seasons also allows teams to analyze the overall trend of player efficiency in different player positions. For example, certain position groups may improve their efficiency over time, whereas others may worsen.

Position-level analysis in basketball frequently faces the challenge of establishing clear definitions of player roles (Beckler et al. 2013; Page et al. 2013). Traditionally, a basketball game employs five player positions: center, power forward, small forward, shooting guard, and point guard (Anıl Duman et al. 2021; McBasketball 2017). Yet, as the game of basketball continues to evolve, the definitions of player positions become less clear, as players with the same traditional position now can play vastly different roles on the court (Whitehead 2019). For example, some point guards may prioritize orchestrating the offense, whereas others may focus on scoring. To better evaluate positional plus-minus, this study redefines eight offensive and three defensive positions. This is achieved through the utilization of

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the Fuzzy K-means algorithm, coupled with play type and official player position data from the NBA (Mallepalle et al. 2020).

Overall, our study employs a hierarchical Bayesian linear model to estimate positional offensive and defensive plus-minus in the NBA from 2015–16 to 2021–22 based on redefined player positions. Then, we compare estimated plus-minus ratings across positions and seasons to determine which positions yield the highest offensive and defensive values on average.

In so doing, our study offers several contributions. First, we present a novel perspective on the values of positions in basketball. Instead of clustering players based on traditional positions, we rely on play type data to inform players' offensive roles on the court (Whitehead 2019). Then, the plus-minus framework with a hierarchical linear model is employed to derive positional values. This modeling structure takes the performance of teammates and opponents into account while maintaining the flexibility of constructing group level parameters (Sabin 2021). Second, we adopt a Bayesian approach to estimate positional plus-minus. Bayesian analysis allows us to quantify the uncertainty of group parameters so inference can be done easily compared to frequentist methods (Jensen et al. 2009; Santos-Fernandez et al. 2019; Thomas et al. 2013). Lastly, we improve on previous player plus-minus ratings by showing that the new player plus-minus ratings produce more predictive results when factoring in player positions.

The rest of this paper is organized as follows. First, we introduce data and models in Sections 2 and 3. We then report the main findings in Section 4. Section 5 discusses possible explanations for these findings. We perform additional analyses and report analysis results in Section 6. Section 7 concludes this paper with a discussion of limitations and future studies.

2 Data

Our analysis of positional plus-minus in the NBA uses data from three main sources. First, we rely on NBA play type data to determine players' offensive positions, and utilize official player position data to identify defensive positions. Play type data document how players use their offensive possessions (McBasketball 2017). Specifically, the company Synergy Sports categorizes every offensive possession into one of the 11 play types (see Table 1). The raw play type data set tracks the number of offensive possessions finished under each play type. We standardize these numbers by computing the percentage of play types for each player. Play

Table 1: Definitions of play types.

Play type	Description
Cut	Players using cut actions to score
Miscellaneous	Possessions do not fit in any other play types
Post-up	Players receiving the ball in the post to score
P & R man	Pick-and-Roll actions with screeners as scorers
Putback	Scoring after receiving offensive rebounds
Transition	Scoring in transition when defense is not set.
Hand-off	Scoring through hand-off actions
Off-screen	Players running off of screens to score
P & R handler	Pick-and-Roll actions with ball handlers as scorers
Spot-up	Players spotting up to score
Isolation	One-on-one actions to score

type data are available on the NBA official statistics website since the 2015–16 season.

It is worth noting that Synergy does not report play types for all players on the NBA website. To be qualified, a player must play at least 10 minutes per game during a regular season and have at least 10 possessions in a particular play type. For example, if a player averaged 5 minutes per game throughout a regular season, this player would not appear in any play type data set. In other cases, if a player averaged more than 10 minutes per game but played fewer than 10 possessions in one play type, this player would not appear in that particular play type data set. Therefore, there exist some missing values in play type data.

We take several steps to handle the missing values to ensure the completeness of the data set. First, for players who play less than 10 minutes per game in a regular season and consequently lack play type data, we categorize them into a new group labeled as low usage players. Table 2 shows the number of low usage players from 2015–16 to 2021–22. On average, low usage players account for 20 percent of all players in a given season. Second, for players missing particular play type data due to the insufficient count of possessions, we assume equal percentages for these missing play types. For example, if a player misses data on three play types with 3 percent of unassigned offensive possessions, then we assume each missing play type accounts for 1 percent of the player's offensive possessions.

We also draw official player position data from the NBA official statistics website in order to identify players' defensive positions. NBA players may play different positions throughout a season. Thus, it is common to see multiple positions associated with one player. The NBA offers seven combinations of player positions between C (Center), F (Forward), and G (Guard) (see Table 3). We rely on these official player positions to group players into one of the three defensive positions, Big, Forward, and Guard.

Table 2: Number of players by position and season.

Position	2015–16	2016–17	2017–18	2018–19	2020–21	2021–22	Total
Skilled Big	44	41	48	46	57	65	301
Versatile Scorer	49	48	52	50	36	44	279
Secondary Attacker	53	45	63	56	85	82	384
Pick and Roll Attacker	65	73	75	75	70	66	424
Roll and Cut Big	52	50	51	45	51	57	306
Movement Shooter	34	42	38	41	28	28	211
Post-up Big	56	41	37	40	29	18	221
Spot-up Shooter	42	49	57	75	88	111	422
Big	98	92	88	87	88	91	544
Forward	126	122	129	137	141	148	803
Guard	171	175	204	204	215	232	1201
Low usage	81	97	120	103	97	134	632

Table 3: Defensive position map.

Defensive position	NBA official player position
Big	C, C–F, F–C
Forward	F, F–G
Guard	G, G–F

Big players are often interior defenders, primarily tasked with protecting the paint area, while Guards and Forwards are generally exterior defenders. We rely on players' primary position, typically denoted by the first letter of their official player positions, to determine their defensive roles. Specifically, if the primary position of a player is C, then this player will be placed into the Big group. Players with the primary position F and G will be assigned to the Forward and Guard groups respectively. The only exception is the players with the official position F–C. In this case, we place this set of players in the Big group rather than the Forward group, since they may appear more often as interior defenders with their secondary position being C.

Lastly, we gather NBA play-by-play data between 2015–16 to 2021–22 from the NBA official statistics website. We omit the NBA 2019–20 season from our analysis, since it was interrupted by the COVID-19 pandemic and had many games played at neutral sites. NBA play-by-play data track a sequence of game-related events on the court, capturing information such as team scores, time left, and player or team actions at a particular point in an NBA game. Many advanced player metrics, such as RAPM, are derived from play-by-play data (Engelmann 2017; Franks et al. 2016; Pelechris and Papalexakis 2016).

From NBA play-by-play data, we extract information related to the performance of stints. A stint can be defined as a unique matchup between five offensive players from

Table 4: Number of stints by season.

Season	Number
2015–16	61,502
2016–17	60,902
2017–18	60,548
2018–19	62,133
2020–21	52,784
2021–22	61,271

one team and five defensive players from another team. We treat a stint as the unit of observation in our statistical model. For each stint in a season, we count the total number of offensive possessions played and the total number of points scored by players on the offensive team. Table 4 reports the total number of unique stints in each season from 2015–16 to 2021–22. It is worth mentioning that some plus-minus studies do not distinguish offensive and defensive teams in stints. Consequently, their counts of stints can be considerably smaller than what we present in Table 4 (Deshpande and Jensen 2016; Jacobs 2019).

3 Methodology

Our analysis of positional plus-minus in the NBA consists of two main steps. First, we establish players' offensive positions by analyzing NBA play type data with the Fuzzy K-means clustering algorithm. Subsequently, we construct a hierarchical linear model to estimate positional plus-minus based on position groups defined in the first step.

The Fuzzy K-means clustering method is a powerful algorithm for partitioning data points into smaller groups. Unlike the traditional K-means algorithm that assigns each data point to a group in a binary way, Fuzzy K-means calculates to which degree each observation belongs to different groups (Bezdek et al. 1984). This approach works well in our analysis, as some players may not perfectly fit into any single position. Thus, we use a player's probabilities of belonging to various

offensive positions to represent the uncertainty associated with Fuzzy K-means clustering results. We also employ the K-means++ algorithm to choose the initial values of centroids that are well separated from each other. This initialization step ensures more stable clustering outcomes (Arthur and Vassilvitskii 2007).

Before implementing the Fuzzy K-means algorithm, we standardize all play type percentage numbers and exclude data from the miscellaneous category. This category, as defined by Synergy Sports, encompasses offensive possessions that cannot be clearly classified into any other play type. In total, we feed 10 standardized play types to the Fuzzy K-means algorithm to group players into distinct offensive roles. For defensive positions, we rely on official player positions to define three defensive roles, Big, Forward, and Guard, as detailed in Section 2.

After assigning one offensive and one defensive position to each player, we propose a hierarchical linear model to estimate positional offensive and defensive plus-minus in the NBA. The model has the following specification:

$$y = \beta_0 + X\beta + Z\gamma + e \quad (1)$$

where X is an m by $2n$ player matrix filled with values of 1, -1 , and 0. The letter n represents the number of unique players in a season. The matrix X has $2n$ columns because the first n columns represent offensive players while the last n columns indicate defensive players. The letter m denotes the total number of stints in a given season. The letter e is a vector of independent errors with a mean equal to 0 and variance equal to σ^2 .

The value of 1 in X indicates a player is on the floor and is with the team playing offense in a stint. The value of -1 suggests a player is on the court and is with the team playing defense. The value of 0 means the player is not on the floor. The letter Z represents an m by $2k$ team matrix filled with 1, -1 , and 0, where k equals the number of unique teams in a season. The first k columns represent offensive teams and the last k columns denote defensive teams. Similar to the player matrix, the values of 1, -1 , and 0 in the Z matrix indicate offensive, defensive, and other teams respectively in a stint.

The response variable y is an m by one vector, representing points scored per 100 possessions by offensive teams in stints. For example, if an offensive team in a stint scored 20 points in 20 possessions, the response value will be equal to 100. Let y_i denotes the i th element of y . Model (2) implies that y_i 's are independent of each other and follow a normal distribution with constant variance, conditional on X^i and Z^i ,

$$y_i | X^i, Z^i \sim N(\beta_0 + X^i\beta + Z^i\gamma, \sigma^2) \quad (2)$$

The intercept term β_0 represents a baseline effect. The coefficient β is a vector of $2n$ player effects. The coefficient γ is a vector of $2k$ team effects. We put the following priors on model coefficients β and β_0 .

$$\begin{aligned} \beta_p &\sim \sum_{j=1}^{11} \omega_j^p N(\mu_j, \tau_j^2), \text{ where player } p \text{ is not low usage} \\ \beta_{low} &\sim N(0, 5), \text{ where player } low \text{ is low usage} \\ \beta_0 &\sim N(0, 1000000) \end{aligned} \quad (3)$$

where the player effect β_p follows a mixture of normal distributions with each normal distribution having a group mean of μ_j and group variance of τ_j^2 , where j is from 1 to 11 with eight offensive and three defensive positions. The notation ω_j^p refers to player p 's probability of belonging to position j , which is estimated from the Fuzzy K-means algorithm for offensive positions. If a player is low usage, then we

assume its player effect β_{low} follows a normal distribution with a mean equal to 0 and variance equal to 5, where *low* refers to a low usage player. The intercept term β_0 has a non-informative normal distribution with mean 0 and variance 1,000,000.

We assume the team effect γ_t comes from a normal distribution with a mean equal to 0 and variance equal to 5.

$$\gamma_t \sim N(0, 5) \quad (4)$$

where t represents an individual team.

Below are the hyperprior distributions for the group mean μ_j , group variance τ_j^2 , error variance σ^2 .

$$\begin{aligned} \mu_j &\sim N(0, 100) \\ \tau_j^2 &\sim IG(0.01, 0.01) \\ \sigma^2 &\sim IG(0.01, 0.01) \end{aligned} \quad (5)$$

We then construct a Gibbs sampler to derive the posterior distributions for group level parameters (i.e., μ_j and τ_j^2). The group level parameters are key of interest, but similar inference can be done for individual level and team level parameters in this study. In Sections 4.2 and 4.3, we will report positional plus-minus and the top player plus-minus ratings for each position group respectively. The details of the Gibbs sampler can be found in Appendix A.

To demonstrate the predictive power of new plus-minus ratings, we also conduct a cross-validation analysis to compare the prediction accuracy of our player plus-minus to other plus-minus metrics. Specifically, we apply 5-fold cross-validation to single season data, and evaluate the Root Mean Square Error (RMSE) between our model and other models. When selecting reference groups for comparison, we pick the RAPM model, which is a non-positional metric, and the model using three position groups (i.e., Big, Forward, Guard) to classify players' offensive and defensive roles. All cross-validation models include an intercept term, player effects, and team effects.

4 Results

4.1 Player positions

We apply the Fuzzy K-mean algorithm to NBA play type data to derive eight clusters, with each cluster representing a distinct offensive player position. The elbow plot (Figure 1) shows the relationship between the number of clusters and the corresponding total within-clusters sum of squares (WSS). Our decision to choose eight clusters is grounded in the examination of the Fuzzy K-means elbow plot and our domain knowledge in basketball. Determining the optimal number of clusters in Fuzzy K-means can be challenging, as the algorithm is often considered as a heuristic method (Brusco and Cradit 2001). Consequently, scholars often search for the stopping point at which the subsequent number of clusters no longer significantly reduces the WSS. In our case, the eighth cluster seems to produce the relatively low WSS. Another consideration of selecting eight clusters is that the chosen number of clusters can result in

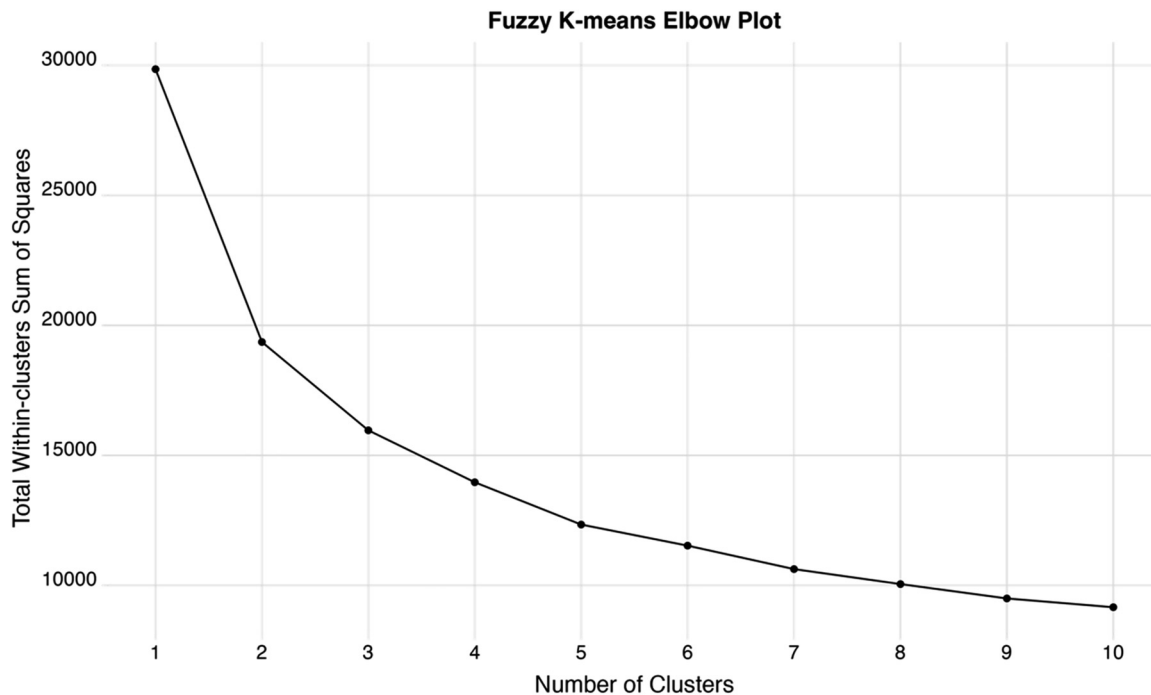


Figure 1: Fuzzy K-means elbow plot.

a sufficient number of players in each cluster (see Table 2). This, in turn, enables player plus-minus to effectively inform positional plus-minus.

In Table 5, we present the centroids of the eight offensive positions. These centroid values serve as the basis for assigning names to the derived positions. These offensive positions are labeled as follows: Roll and Cut Big, Post-up Big, Skilled Big, Movement Shooter, Spot-up Shooter, Versatile Scorer, Pick and Roll Attacker, and Secondary Attacker. In particular, players categorized as Roll and Cut Bigs often employ pick-and-roll and cutting actions as their primary means of scoring. They also contribute to team offense through putbacks after securing offensive rebounds. Post-up Bigs predominantly depend on post-up actions to score points. They also utilize pick-and-roll actions to score, but

not as frequently as Roll and Cut Bigs. It is worth noting that there is a substantial decline in the presence of Post-up Bigs in the league over time. Table 2 shows the number of Post-up Bigs has reduced from 56 players during the 2015–16 season to a mere 18 players in the 2021–22 season. On the other hand, Skilled Bigs, who can score not only through pick-and-rolls but also spot-up shooting, have witnessed a notable growth in their representation. Their numbers have increased from 44 in 2015–16 to 65 in 2021–22 (see Table 2).

Movement Shooters appear to be the best shooters in the league. Their offensive possessions are largely finished with off-screen, hand-off, and spot-up plays, which require proficient shooting skills in players. In contrast to Movement Shooters, Spot-up Shooters exceedingly rely on spot-up shooting on offense. They often position in the

Table 5: Centroids of play types for offensive positions.

	Cut	Post-up	P & R man	Putback	Transition	Hand-off	Off-screen	P & R handler	Spot-up	Isolation
Skilled Big	10 %	7 %	14 %	8 %	12 %	2 %	3 %	3 %	30 %	4 %
Versatile Scorer	3 %	4 %	2 %	2 %	15 %	5 %	4 %	31 %	15 %	14 %
Secondary Attacker	5 %	2 %	2 %	3 %	19 %	7 %	5 %	18 %	30 %	5 %
Pick and Roll Attacker	3 %	2 %	1 %	2 %	15 %	6 %	3 %	36 %	20 %	7 %
Roll and Cut Big	24 %	6 %	23 %	18 %	8 %	1 %	1 %	1 %	7 %	2 %
Movement shooter	4 %	2 %	2 %	2 %	16 %	11 %	16 %	13 %	28 %	3 %
Post-up Big	13 %	22 %	19 %	11 %	7 %	1 %	2 %	1 %	12 %	4 %
Spot-up Shooter	7 %	3 %	4 %	5 %	19 %	4 %	4 %	7 %	39 %	3 %

Table 6: Mean and standard deviation of positional plus-minus mean estimate.

Position	2015–16	2016–17	2017–18	2018–19	2020–21	2021–22
Skilled Big	2.9 (1.1)	3.2 (1.1)	2.9 (1.2)	2.9 (1.1)	3.6 (1.1)	3.3 (1)
Versatile Scorer	5.8 (1.2)	5.9 (1.3)	4.7 (1.1)	5 (1.1)	7.6 (1.2)	5.7 (1.2)
Secondary Attacker	4 (1)	4 (1.1)	3.2 (1)	3.7 (1)	4.3 (1)	2.9 (1)
Pick and Roll Attacker	4.6 (1)	4.2 (1.2)	3.5 (1.1)	3.7 (1)	4.8 (1.1)	3.2 (1.1)
Roll and Cut Big	1.5 (1)	2.6 (1.1)	0.2 (1.2)	3.4 (1.1)	3.1 (1.2)	4 (1.1)
Movement Shooter	4.8 (1.2)	4.4 (1.3)	4.1 (1.1)	3.9 (1.1)	5.8 (1.3)	3.8 (1.2)
Post-up Big	3.1 (1.1)	3.7 (1.2)	1.6 (1.4)	4.6 (1.2)	3.8 (1.4)	3.9 (1.3)
Spot-up Shooter	3.6 (1.1)	4.1 (1.1)	2.6 (1)	3.7 (1)	3.8 (1)	2.9 (1)
Big	5.1 (1.1)	3.6 (1)	3.3 (1)	3.7 (0.9)	2.4 (1.2)	3 (1.3)
Forward	3.4 (1.1)	2.2 (1)	1.9 (0.8)	1.8 (0.9)	1.3 (1)	1.7 (1.2)
Guard	1.7 (1)	−0.2 (0.9)	1.2 (0.9)	0.1 (0.9)	−0.4 (1)	1.6 (1.2)

Standard deviations are in parentheses.

Table 7: Mean and standard deviation of positional plus-minus variance estimate.

Position	2015–16	2016–17	2017–18	2018–19	2020–21	2021–22
Skilled Big	9.5 (6)	3.1 (3.8)	6.3 (5.4)	8.5 (6.3)	2.8 (3.7)	12.7 (1.8)
Versatile Scorer	10.3 (5.4)	16.9 (6.8)	7.9 (5.2)	8.2 (5.3)	6.6 (5.7)	5.5 (6)
Secondary Attacker	2.3 (2.7)	4.1 (3.8)	2.2 (2.6)	8.7 (4.4)	2.6 (2.8)	9.3 (3.7)
Pick and Roll Attacker	3.4 (3.2)	11.1 (4.6)	4.9 (3.5)	7.6 (4.2)	8 (4.4)	4 (4.5)
Roll and Cut Big	2.3 (3)	1.1 (1.8)	12.3 (6.7)	2.9 (3.8)	3.3 (3.9)	3.7 (4.4)
Movement Shooter	8 (6.2)	11.6 (7.1)	2.5 (3.4)	6.6 (5.4)	4 (4.8)	4.7 (4.7)
Post-up Big	3.2 (3.2)	12.1 (6.5)	20.3 (9.5)	9.3 (6.9)	11 (9.9)	3.1 (5.9)
Spot-up Shooter	2.1 (2.9)	1 (1.6)	2.8 (3.1)	7.3 (4.5)	1.7 (2.4)	5 (2.7)
Big	3.2 (2.7)	1.6 (1.8)	2.6 (2.5)	1.1 (1.5)	0.6 (1)	1.4 (0.9)
Forward	3.1 (2.2)	3.4 (2.4)	0.6 (1)	4.3 (2.3)	2.2 (2.3)	4.4 (1.6)
Guard	4.5 (1.9)	1.8 (1.4)	6.3 (2.1)	2.5 (1.6)	3.1 (2.2)	5 (1.7)

Standard deviations are in parentheses.

corners of a basketball court and wait to catch and shoot the ball. Notably, the number of Spot-up Shooters has increased considerably since the NBA 2020–21 season (see Table 2). Versatile Scorers are all-around players capable of scoring through various actions, such as isolation, transition, and pick-and-roll. Next, Pick-and-Roll Attackers primarily employ pick-and-roll strategies to score points, dedicating far fewer possessions to isolation plays than Versatile Scorers. Lastly, Secondary Attackers are intermediate players positioned between Pick-and-Roll Attackers and Spot-up Shooters. While they can score through pick-and-roll actions, they do not do so as often as Pick-and-Roll Attackers. Similarly, they can score with spot-up shooting, but not as frequently as Spot-up Shooters.

4.2 Positional plus-minus

After we determine offensive and defensive positions for players, we run the same hierarchical Bayesian linear model for each of the six NBA seasons separately to estimate

positional plus-minus. Tables 6 and 7 summarize the mean and standard deviation of position level mean and variance parameters. In the following text, we focus on comparing the values of parameters across positions and seasons as well as interpreting the differences in these values.

4.2.1 Comparison of plus-minus across positions by season

Figure 2 visualizes average offensive plus-minus for various positions in the NBA from the 2015–16 to 2021–22 seasons.¹ These results indicate that Versatile Scorers consistently outperform the other positions in terms of offensive plus-minus. To estimate the probabilities that Versatile Scorers' offensive plus-minus is higher than the others, we draw samples from the posterior distributions of offensive

¹ Dashed lines represent posterior means.

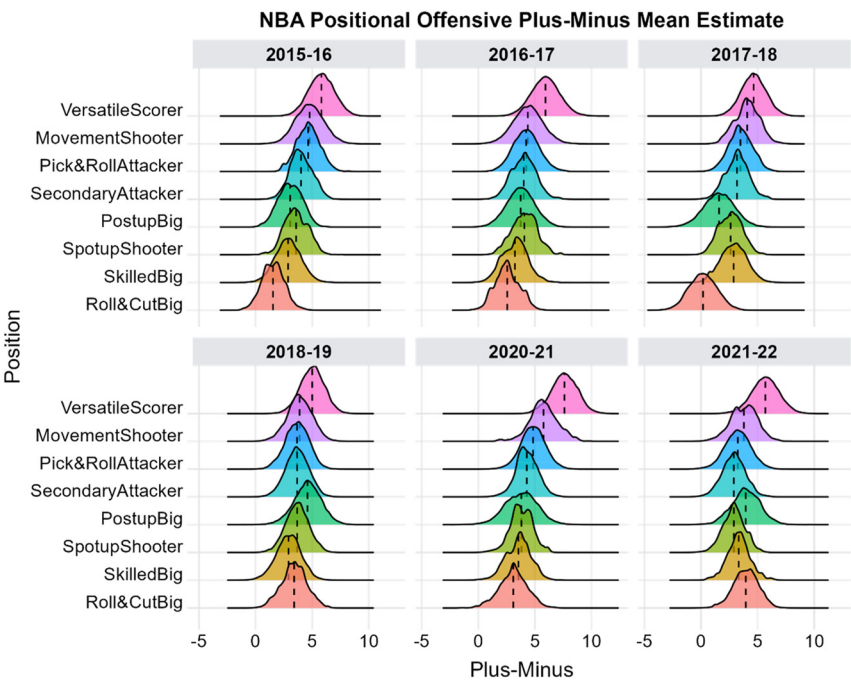


Figure 2: NBA positional offensive plus-minus mean estimate from 2015–16 to 2021–22.

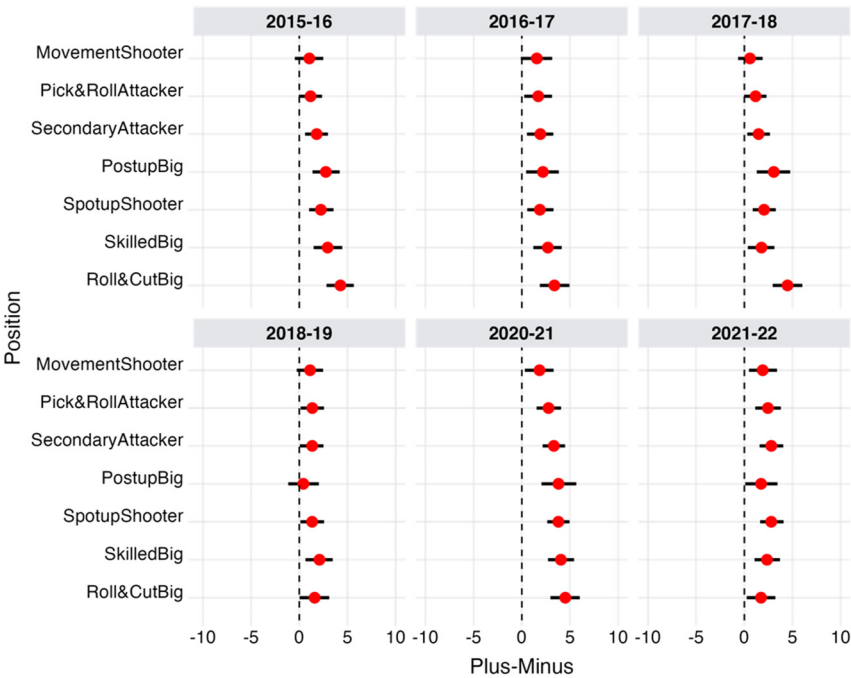


Figure 3: Eighty percent credible intervals of the differences of offensive plus-minus between Versatile Scorers and the other offensive positions.

plus-minus and compute the differences between these posterior samples. Figure 3 displays the 80 % credible intervals of the differences of offensive plus-minus between Versatile Scorers and the other positions. Notably, the majority of these intervals lie above zero, indicating that Versatile

Scorers are the most valuable offensive contributors in most seasons.²

² Red dots represent the means of differences.

Another observation from Figure 2 is that the gaps in positional offensive plus-minus have diminished over the years. In particular, there are notable differences in offensive plus-minus in 2015–16 among position groups, with Versatile Scorers leading and Roll and Cut Bigs at the bottom of the list. By the 2021–22 season, these differences narrowed as the average offensive plus-minus values for each position move closer to each other. However, Versatile Scorers consistently stood out, maintaining higher plus-minus ratings than the rest. This trend suggests that while distinctions in positional offensive plus-minus have become subtler over the years, Versatile Scorers persist as the cornerstone of team offense.

On the defensive side, Figure 4 illustrates that Bigs' average defensive plus-minus consistently ranks the highest among the position groups. To estimate the probabilities that Bigs' defensive plus-minus surpasses the other positions' defensive plus-minus, we construct the 80 % credible intervals of the differences of defensive plus-minus between Bigs and the other defensive positions

from 2015–16 to 2021–22 in Figure 5. Most intervals locate well above zero, providing strong evidence that Bigs create more defensive value than the others in the NBA.

In addition to deriving average positional ratings (μ_j), the variance of positional plus-minus (τ_j^2) can be easily estimated in our study. Figure 6 shows the consistently low variance of offensive plus-minus for Spot-up Shooters compared to the other offensive positions. On the other hand, Post-up Bigs and Versatile Scorers display notably higher variances of their offensive plus-minus relative to the other positions. These observations indicate that there might be significant variation in offensive plus-minus among players in these two positions. Meanwhile, the offensive efficiency of Spot-up Shooters appears to be relatively consistent across different players.

Figure 7 plots the posterior distributions of positional defensive plus-minus variance estimates. The mean variance of Guards' defensive plus-minus seems notably greater than these of the other positions. This finding indicates that Guards display a broader range of defensive efficiency

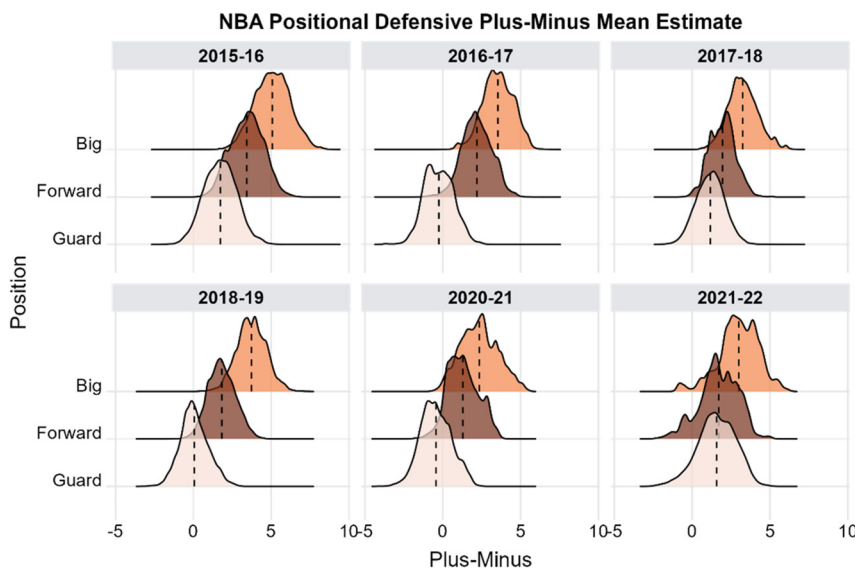


Figure 4: NBA positional defensive plus-minus mean estimate from 2015–16 to 2021–22.

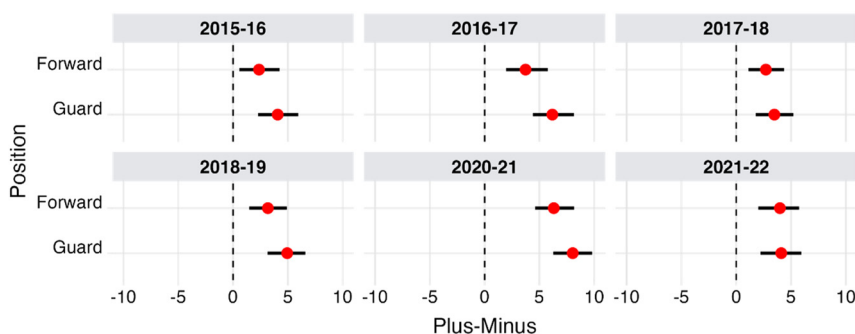


Figure 5: Eighty percent credible intervals of the differences of defensive plus-minus between Bigs and the other defensive positions.

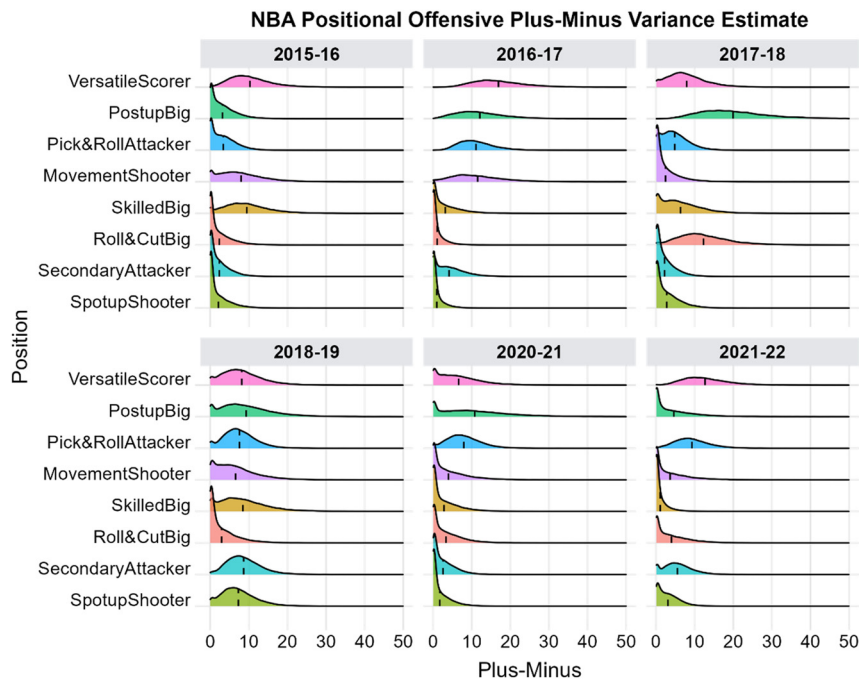


Figure 6: NBA positional offensive plus-minus variance estimate from 2015–16 to 2021–22.

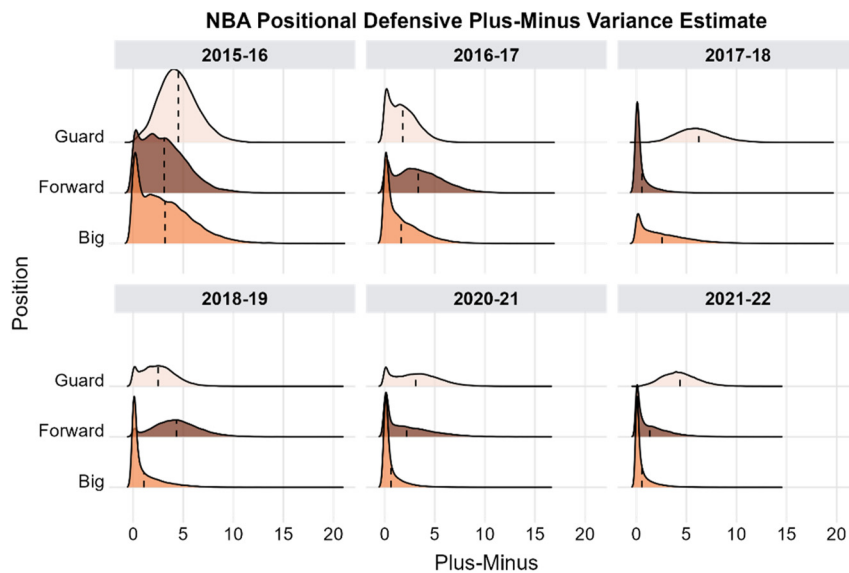


Figure 7: NBA positional defensive plus-minus variance estimate from 2015–16 to 2021–22.

compared to Forwards and Bigs. The 80 % credible intervals of the variance differences between Guards and the other defensive positions do not include zero in most cases.³ This offers some evidence that the variation in Guards' defensive plus-minus is greater than the other positions'.

³ Eighty percent credible intervals are not shown here. They are available upon request.

4.2.2 Comparison of plus-minus across seasons by position

Figure 8 aims to compare offensive plus-minus across seasons by position. In particular, Roll and Cut Bigs have shown significant progress in their offensive plus-minus from 2015–16 to 2021–22. Both Post-up Bigs and Skilled Bigs also saw enhancements in their offensive plus-minus over

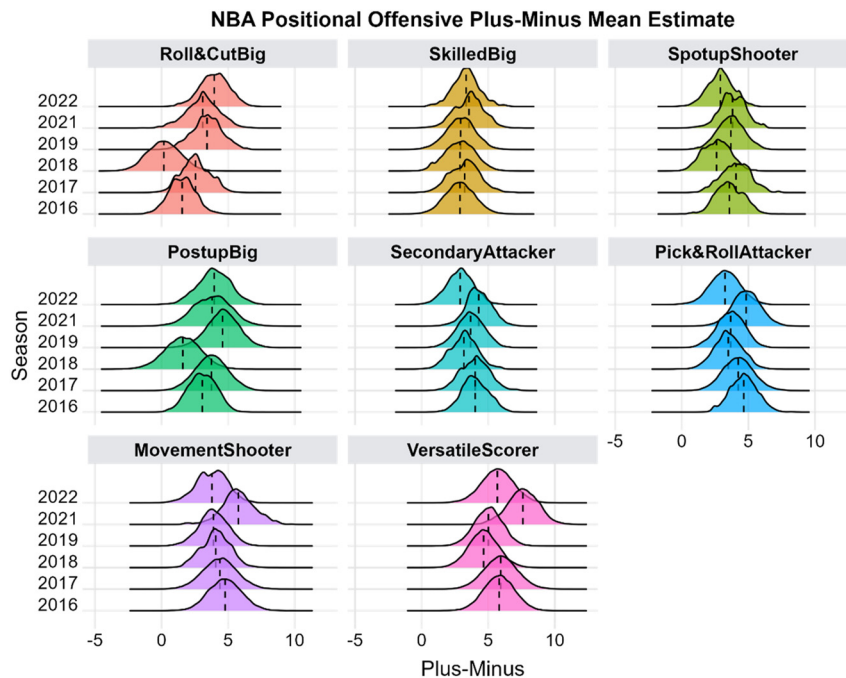


Figure 8: NBA offensive plus-minus mean estimate by position.

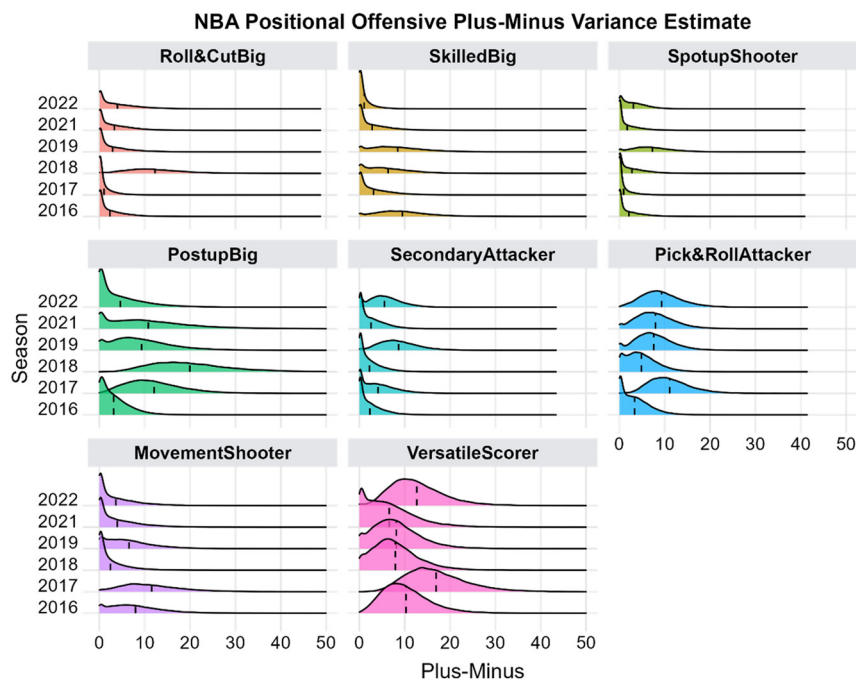


Figure 9: NBA offensive plus-minus variance estimate by position.

this period. The other positions' offensive plus-minus stays relatively stable from season to season.

Figure 9 shows the posterior distributions of offensive plus-minus variance estimates across eight offensive roles. The mean variances of positional offensive plus-minus

appear to be fairly consistent over the seasons. Figures 10 and 11 visualize the posterior distributions of defensive plus-minus mean and variance estimates respectively. Neither of them exhibits any particular pattern over the years.

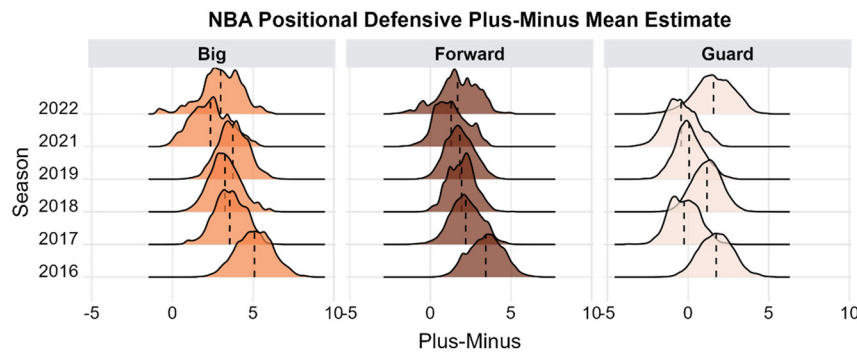


Figure 10: NBA defensive plus-minus mean estimate by position.

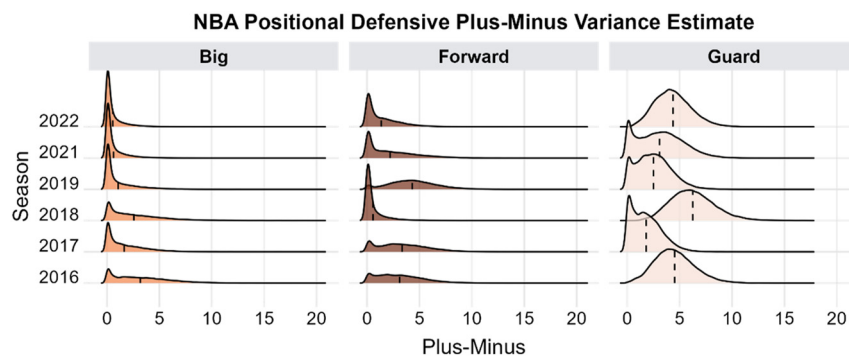


Figure 11: NBA defensive plus-minus variance estimate by position.

4.3 Player plus-minus and cross-validation

While our analysis of plus-minus focuses on estimating positional values in the NBA, it maintains the flexibility of estimating plus-minus ratings for individual players. We report the top five plus-minus ratings for each offensive and defensive position group in Appendix C.

We also conduct cross-validation tests to demonstrate the improved predictive power of our player plus-minus ratings. Table 8 shows the RMSE for RAPM with a range of penalty values λ (i.e., 500, 1500, and 3000), the RMSE for the model using three traditional basketball position groups (3-position) (i.e., Big, Forward, Guard), and the RMSE for our model with redefined eight offensive positions (8-position). Cross-validation results show both 3-position and 8-position

models outperform the RAPM models in all six seasons. This finding suggests that the player position is an important factor to consider in estimating player plus-minus. In addition, the 8-position model produces the best RMSE among all models across six seasons. This result indicates that redefined offensive positions by play type data may better capture offensive roles of NBA players than traditional position groups in basketball, and thus can help estimate more predictive player plus-minus ratings.

5 Discussion

Recall in Figures 2 and 3 we show that Versatile Scorers produce the highest offensive value among all the offensive

Table 8: Cross-validation results.

Model	λ	2015–16 RMSE	2016–17 RMSE	2017–18 RMSE	2018–19 RMSE	2020–21 RMSE	2021–22 RMSE
RAPM	500	78.725	79.758	80.152	79.921	80.223	80.612
RAPM	1500	78.710	79.739	80.142	79.911	80.207	80.610
RAPM	3000	78.733	79.763	80.171	79.940	80.232	80.626
3-position		78.691	79.718	80.118	79.900	80.194	80.600
8-position		78.685	79.701	80.118	79.899	80.181	80.590

Bold values represent the RMSE from the model adopted in the present study. Lower RMSE values indicate better model performance.

positions. We define Versatile Scorers as players who primarily rely on isolation, transition, and pick-and-roll actions to score points. As such, these players are likely to be the ball-dominant players who control the ball most of the time on the court. Teams often allow star players to take on this role because they are generally more proficient at scoring. Hence, it is not surprising that Versatile Scorers register higher offensive plus-minus ratings than players in the other positions.

By comparing estimation results across the six NBA seasons, we show that the gaps in offensive plus-minus between the eight offensive positions have diminished over time. After further examining how offensive plus-minus have changed over the years for each position, we notice that the improvement of Roll and Cut Bigs may significantly contribute to the shortened gaps. Specifically, Figure 8 shows a consistent upward trend in their offensive plus-minus from 2015–16 to 2021–22. Considering that the quantity of Roll and Cut Bigs remains relatively stable over the years (as shown in Table 2), we believe the enhanced offensive plus-minus could be attributed to increased offensive efficiency among Roll and Cut Bigs. Furthermore, we note a gradual increase in Post-up Bigs' and Skilled Bigs' offensive plus-minus over time, though their plus-minus ratings do not seem to rise as significantly as Roll and Cut Bigs'.

The estimated variances of positional plus-minus also present some interesting findings. Recall in Figure 6 that the mean variances of Spot-up Shooters' offensive plus-minus are the smallest in most seasons spanning from 2015–16 to 2021–2022. This finding could be attributed to the role of Spot-up Shooters, which typically involves less prominent involvement in team offense. On the contrary, the mean variances of Post-up Bigs' and Versatile Scorers' offensive plus-minus are consistently higher than the others in Figure 6. The steep decline in quantity and the relatively low number of Post-up Bigs in the NBA may partially explain the high variation in Post-up Bigs' offensive plus-minus (see Table 2). This finding may also indicate that scoring proficiency in offense-oriented positions, such as Versatile Scorers and Post-up Bigs, can vary greatly from one player to another in the NBA.

On the defensive side, Figure 4 shows that Bigs tend to provide more defensive value than the other positions. This finding is not surprising as Bigs are not only responsible for defending opposing Bigs but also play a crucial role in preventing opponents from scoring in the paint area. As such, interior players may carry more defensive responsibility, creating more defensive impact than exterior players. Guards, limited by their size, offer the least

defensive value to teams in Figure 4. Additionally, it is worth noting that despite Guards having the lowest defensive plus-minus, they exhibit the highest variability in defensive performance among the three defensive positions. In simpler terms, the defensive plus-minus values for Guards are more dispersed around the average compared to those of Bigs and Forwards. This observation implies that there may exist some exceptional defenders or some extremely poor defenders in the Guard position.

Another significant contribution of our findings is that the prediction accuracy of the new player plus-minus surpasses that of many existing player plus-minus metrics, as reported in Section 4.3. While our study does not intend to optimize the prediction accuracy of player effects, it shows that factoring in player positions when setting priors for player plus-minus may improve metric performance (Matano et al. 2023). A similar study conducted by Intraocular (2022) echoes our finding that considering player positions in player prior distributions may help produce more predictive player plus-minus. However, one major difference between our analysis and Intraocular's model lies in the fact we assign players into one of the eight offensive roles based on their play types, while Intraocular's model considers traditional basketball player positions, Bigs, Wings, and Guards. Our cross-validation results in Table 8 suggest that player plus-minus estimated from the 8-position model may perform better than the estimates from the 3-position model in most seasons. This finding suggests that using well-defined player roles can potentially enhance the predictiveness of player plus-minus ratings.

Our analysis of positional plus-minus also resembles Sabin's (2021) study that uses similar Hierarchical Bayesian models to estimate player plus-minus in the NFL. In constructing priors for player-level plus-minus, he redefined variance parameters in a t-distribution by linking them to position-level parameters. Our study follows a similar hierarchical structure in constructing priors for player-level plus-minus, but with different prior distributions (i.e., a mixture of Gaussians). Additionally, Sabin (2021) considered linear weighted player positions in constructing prior parameters with weights being players' predicted percentages of snaps at different positions. Our model also uses weighted positions in player priors, but with the weights determined by the probabilities of a player belonging to each position group. Despite these differences, both our model and Sabin's model point to the same conclusion that factoring in player positions produces more predictive player plus-minus results.

6 Additional analyses

To check the robustness of our estimation results, we conduct a few additional analyses. First, we refine our play-by-play data set by filtering out garbage time in which teams do not place their best possible players on the court. Second, we re-estimate both positional offensive and defensive plus-minus based on traditional basketball player positions, Guard, Forward, and Big. We report these analysis results below.

6.1 Garbage time

We employ the definition of garbage time from the basketball analytics website, *Cleaning the Glass* (Falk n.d.). In general, the website considers the time when the score differential is large toward the end of the game and two teams no longer field their best players as garbage time. Specifically, three factors are used to determine garbage time, the quarter, score differential, and the number of starters on the floor. For example, if a game is in the fourth quarter, one team is leading the other by 15 points with 2 min left on the clock, and the total number of starters on the floor combined from the two teams is 1, then the website will consider this period as garbage time. The details of garbage time thresholds can be found in Table 9 in Appendix B.

Overall, we find similar positional plus-minus estimation results as before. To better visualize these findings,

we place the estimates from play-by-play data with and without garbage time side by side in Figures 12 and 13. It is clear that removing garbage time from play-by-play data does not significantly alter our prior findings, except for the 2020–21 season when positional plus-minus from the full data set seems higher than positional plus-minus derived from the data set without garbage time. We suspect this is because 2020–21 season games contain more garbage time than games in the other seasons, as 2020–21 is the first full season since the interrupted 2019–20 season due to the COVID-19 pandemic. Data analysis provides some support for this possibility. After counting the percentage of garbage time in each of the six seasons examined in this study, it appears that almost 4 percent of possessions are classified as garbage time in the 2020–21 season, whereas only 3.4 to 3.7 percent of possessions are considered garbage time in the other seasons.

6.2 Three offensive positions

We also re-estimate positional offensive plus-minus based on three positions groups (i.e., Guard, Forward, and Big). Recall that our main analysis defines eight offensive roles based on players' play types, while defensive positions are derived from traditional basketball positions. Here we re-estimate offensive plus-minus similarly to defensive plus-minus in order to compare the results derived from conventional player positions with those from play type-based positions.

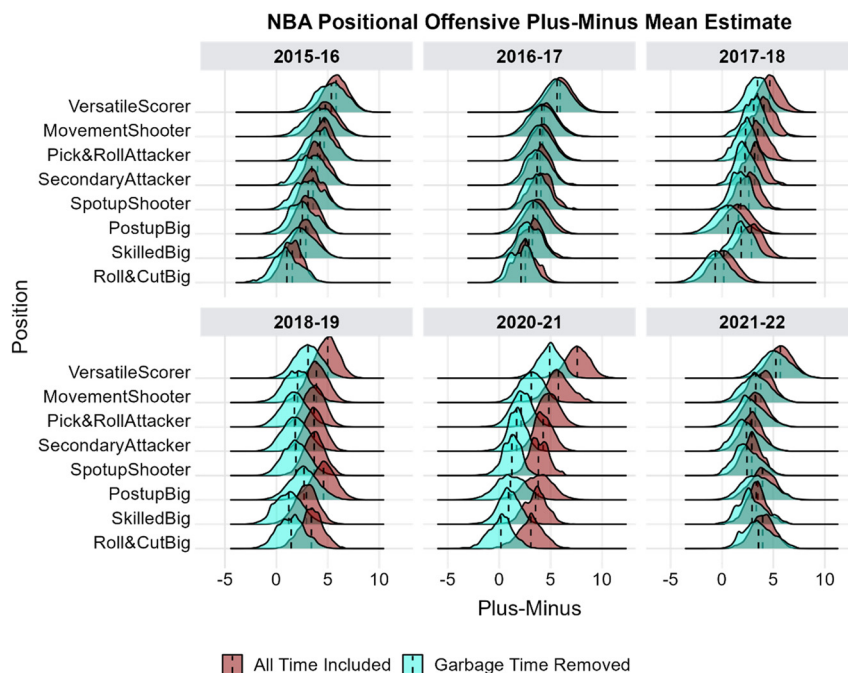


Figure 12: Comparison of positional offensive plus-minus mean estimate across seasons with and without garbage time.

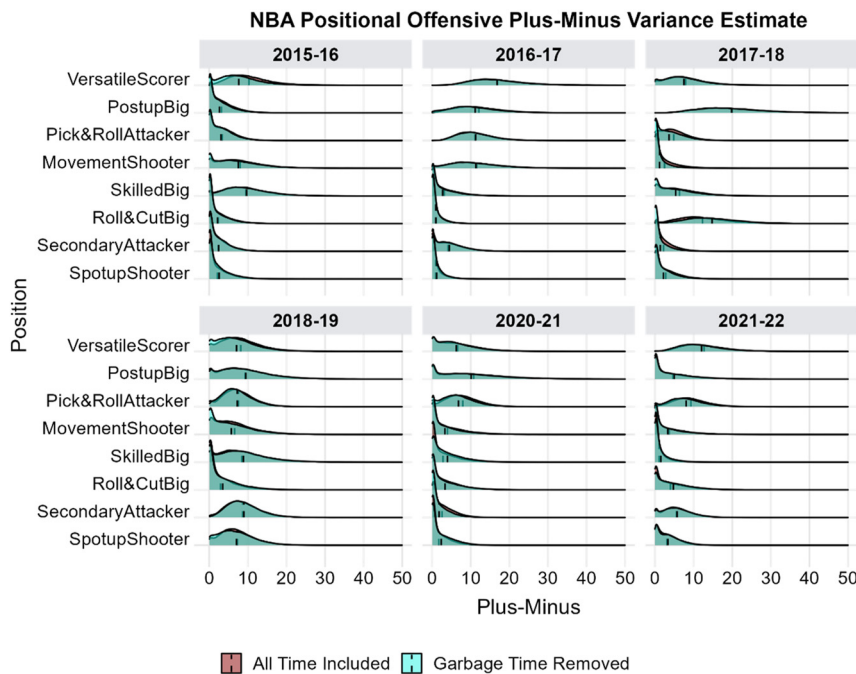


Figure 13: Comparison of positional offensive plus-minus variance estimate across seasons with and without garbage time.

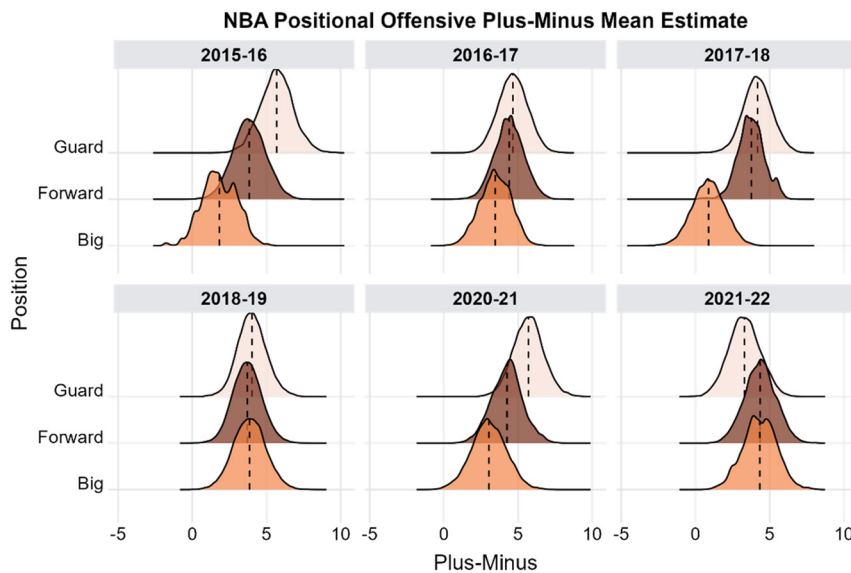


Figure 14: NBA offensive plus-minus from 2015–16 to 2021–22 with three offensive positions.

Figures 14 and 15 show the posterior distributions of offensive plus-minus for the three traditional positions, Guard, Forward, and Big. Similar to our main findings, the estimates suggest the gaps in positional offensive plus-minus have shrunk over time. In particular, Bigs' offensive plus-minus has increased considerably from 2015–16 to 2021–22. These findings provide further evidence that the improvement by Bigs narrows the gap in offensive plus-minus between the eight offensive positions.

Figure 16 compares offensive and defensive plus-minus variance estimates between Bigs, Forwards, and Guards. Results show that the mean variances of offensive plus-minus generally surpass those of defensive plus-minus in most seasons. This suggests that the variation in defensive ability is relatively small among NBA players than offensive ability. Figure 16 also shows the low correlations between offensive and defensive plus-minus variance estimates.

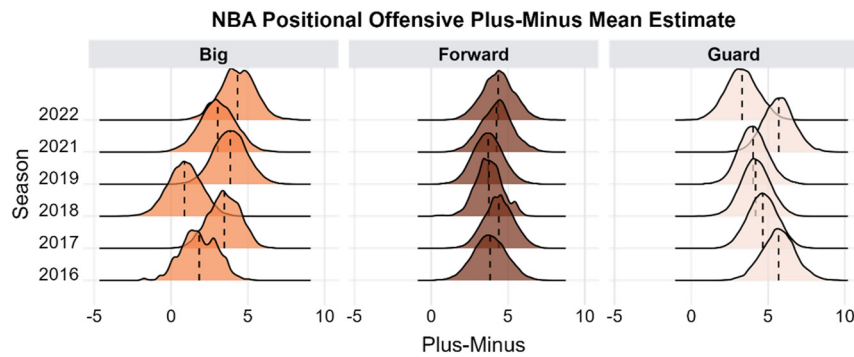


Figure 15: NBA offensive plus-minus across three offensive positions.

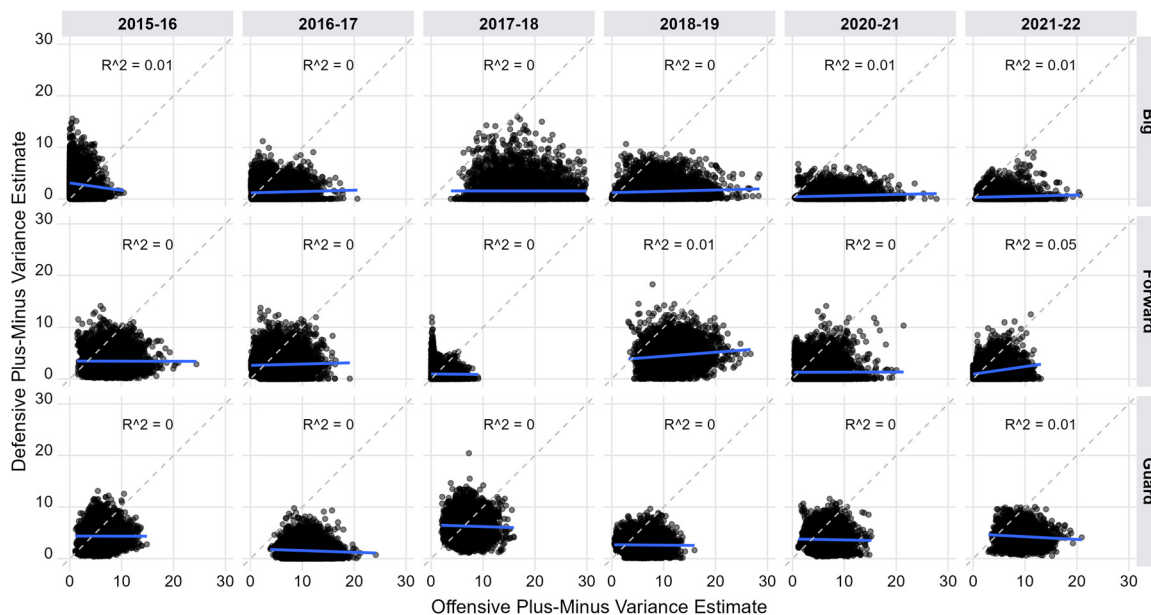


Figure 16: Comparison of NBA positional plus-minus variance estimate between Bigs, Forwards, and Guards.

7 Conclusions

In this study, we have employed a hierarchical Bayesian linear model to estimate positional plus-minus from 2015–16 to 2021–22 based on redefined offensive and defensive positions in the NBA. Our analysis finds that Versatile Scorers and Bigs have the highest positional offensive and defensive plus-minus respectively. We also show that the variance of Spot-up Shooters' offensive plus-minus is the lowest among the eight offensive positions, while the variances of Post-up Bigs' and Versatile Scorers' offensive plus-minus are substantially higher than the others. On the defensive side, we note that Guards have the lowest average defensive plus-minus, but the highest variance among the three defensive positions.

Through the analysis of positional offensive plus-minus from 2015–16 to 2021–22 in the NBA, it becomes evident

that the gaps in offensive plus-minus between the eight offensive positions have decreased over the years. That is, the positional offensive plus-minus values become closer to each other over time, except for Versatile Scorers, whose offensive plus-minus consistently ranks higher than the others. Additionally, we have noted that the increase in offensive plus-minus for Roll and Cut Bigs may play a significant role in closing the gaps.

In addition to estimating positional plus-minus values and comparing them across position groups and seasons, our analysis develops more predictive plus-minus ratings for individual NBA players. Cross-validation results show that the new player plus-minus produces higher prediction accuracy than non-positional plus-minus models (i.e., RAPM) and the 3-position plus-minus model in most seasons.

Our study of positional plus-minus has a few limitations. First, the response variable, points scored per 100

possessions, employed in this study does not reflect the importance of points in a game. In fact, we treat points scored or allowed equally throughout a game. However, it is common to see points scored in a close game toward the end of the game are far more important than points acquired in the first quarter. Second, the clustering results for offensive positions are subjective as the Fuzzy K-means algorithm is a heuristic method. Therefore, the eight offensive positions we proposed in this study may not be the best solution for grouping players.

Future studies of positional plus-minus may consider using win probability as the response variable. The calculation of win probability takes game situations into consideration. Thus, regressing the change in win probability over stints may better estimate positional contributions to team success (Deshpande and Jensen 2016). Future research may also employ new data to cluster players. In particular, the availability of player tracking data in major sports leagues may allow scholars to produce more detailed and accurate classifications of player positions in sports.

Acknowledgments: The authors would like to thank the editors and reviewers for their constructive and thoughtful feedback and comments.

Research ethics: Not applicable.

Author contributions: The authors have accepted responsibility for the entire content of this manuscript and approved its submission.

Competing interests: The authors state no conflict of interest.

Research funding: None declared.

Data availability: Not applicable.

Appendix A

We construct a hierarchical Bayesian liner model to estimate positional plus-minus in this study. The Gibbs sampling method with 30,000 iterations and a burn-in period of 1000 is used to derive the estimates in the model. Our variables of interest are the mean and variance of positional plus-minus μ_j and τ_j^2 , where j is from 1 to 11 with eight offensive positions and three defensive positions.

To facilitate Gibbs sampling, we introduce a vector of latent variables z^p to the mixture of normal distributions

in Equation (3). The vector indicates which position group player p belongs to in an iteration and has a multinomial distribution with the player's probabilities of belonging to different position groups. For each iteration in Gibbs sample, we first draw z for each player. Then we use the following conditional posterior distributions for μ_j and τ_j^2 to draw posterior samples.

$$z^p \propto \text{Multinomial}(\omega_1^p, \omega_2^p, \dots, \omega_{11}^p) \quad (6)$$

$$p(\mu_j | z^p = j) \propto N \left(\left(k + \frac{\tau_j^2}{100} \right)^{-1} \left(\sum \beta_{p_j} \right), \right. \\ \left. \tau_j^2 \left(k + \frac{\tau_j^2}{100} \right)^{-1} \right) \quad (7)$$

$$p(\tau_j^2 | z^p = j) \propto IG \left(\frac{k}{2} + 0.01, 0.5 * \sum (\beta_{p_j} - \mu_j)^2 + 0.01 \right) \quad (8)$$

where k is the number of players that belong to group j in an iteration. The notation p_j refers to player p who belongs to position j in an iteration. The letter β represents player effects.

Appendix B

Table 9: Thresholds of NBA garbage time defined by the website Cleaning the Glass.

Condition	
1	Game is in the 4th quarter
2	Score differential is larger than 25 when 9–12 min left
3	Score differential is larger than 20 when 6–9 min left
4	Score differential is larger than 10 when 0–6 min left
5	A total of two or fewer starters combined from the two teams on the field

The website also notes the game can never change from being garbage time to being non-garbage time. If any of the above conditions is not met and the game switches from being garbage time to being non-garbage time, then the garbage time clock resets.

Appendix C

Table 10: NBA 2015–16 top five plus-minus ratings by position.

Player	Offensive plus-minus	Position	Position probability
Draymond Green	8.2	Skilled Big	68 %
Patrick Patterson	5.9	Skilled Big	47 %
Mirza Teletovic	5.7	Skilled Big	27 %
Serge Ibaka	5.2	Skilled Big	76 %
Kristaps Porzingis	5.1	Skilled Big	53 %
LeBron James	10.2	Versatile Scorer	72 %
Kevin Durant	8.7	Versatile Scorer	49 %
Chris Paul	8.7	Versatile Scorer	71 %
Russell Westbrook	8.6	Versatile Scorer	60 %
James Harden	7.9	Versatile Scorer	71 %
Eric Gordon	5.4	Secondary Attacker	42 %
Anthony Morrow	5.1	Secondary Attacker	30 %
Iman Shumpert	5	Secondary Attacker	43 %
Kentavious Caldwell-Pope	4.9	Secondary Attacker	79 %
C. J. Watson	4.7	Secondary Attacker	40 %
Kyle Lowry	8.1	Pick & roll Attacker	56 %
Jrue Holiday	6.5	Pick & roll Attacker	65 %
Andre Miller	6	Pick & roll Attacker	41 %
Ricky Rubio	6	Pick & roll Attacker	85 %
Rodney Hood	5.9	Pick & roll Attacker	44 %
Brandon Bass	3.7	Roll & cut Big	48 %
Willie Cauley-Stein	2.9	Roll & cut Big	80 %
Amir Johnson	2.7	Roll & cut Big	78 %
Alex Stephenson	2.6	Roll & cut Big	27 %
Cody Zeller	2.4	Roll & cut Big	70 %
Avery Bradley	7	Movement Shooter	83 %
Wesley Matthews	6.9	Movement Shooter	33 %
Klay Thompson	6.9	Movement Shooter	52 %
J. J. Redick	6.5	Movement Shooter	53 %
Nicolas Batum	6.2	Movement Shooter	64 %
Ryan Anderson	5.5	Post-up Big	36 %
Dirk Nowitzki	4.7	Post-up Big	62 %
Gorgui Dieng	4.5	Post-up Big	87 %
Paul Millsap	4.5	Post-up Big	56 %
Karl-Anthony Towns	4.3	Post-up Big	85 %
Jason Terry	4.8	Spot-up Shooter	47 %
Danny Green	4.7	Spot-up Shooter	52 %
Solomon Hill	4.5	Spot-up Shooter	64 %
Axel Toupane	4.5	Spot-up Shooter	66 %
Harrison Barnes	4.4	Spot-up Shooter	33 %
Player	Defensive plus-minus	Position	
Nikola Jokic	6.5	Big	
Steven Adams	6.4	Big	
Tim Duncan	6.3	Big	
DeAndre Jordan	6.2	Big	
Dwight Powell	6.2	Big	

Table 10: (continued)

Player	Defensive plus-minus	Position
Draymond Green	6.3	Forward
Kawhi Leonard	5.3	Forward
Wesley Johnson	4.8	Forward
Blake Griffin	4.7	Forward
Chris Bosh	4.7	Forward
Tony Snell	4.6	Guard
Danny Green	4.5	Guard
Kyle Korver	3.9	Guard
Evan Turner	3.7	Guard
Michael Carter-Williams	3.6	Guard

Table 11: NBA 2016–17 top five plus-minus ratings by position.

Player	Offensive plus-minus	Position	Position probability
Patrick Patterson	5	Skilled Big	45 %
Draymond Green	4.4	Skilled Big	55 %
Marvin Williams	4.2	Skilled Big	73 %
Okaro White	4.2	Skilled Big	48 %
Frank Kaminsky	4.2	Skilled Big	73 %
Kawhi Leonard	11.7	Versatile Scorer	80 %
Kyle Lowry	10.9	Versatile Scorer	61 %
LeBron James	9.8	Versatile Scorer	62 %
Russell Westbrook	9.8	Versatile Scorer	82 %
Chris Paul	9.6	Versatile Scorer	82 %
Patty Mills	6.6	Secondary Attacker	52 %
Tim Hardaway Jr.	5.8	Secondary Attacker	45 %
Aaron Gordon	5.7	Secondary Attacker	45 %
Jae Crowder	5.6	Secondary Attacker	36 %
Norman Powell	5.6	Secondary Attacker	39 %
Isaiah Thomas	10.6	Pick & roll Attacker	42 %
Lou Williams	8.8	Pick & roll Attacker	74 %
Kemba Walker	8.5	Pick & roll Attacker	76 %
Darren Collison	8.3	Pick & roll Attacker	88 %
Mike Conley	8.2	Pick & roll Attacker	46 %
David Lee	3.5	Roll & cut Big	71 %
Mason Plumlee	3.5	Roll & cut Big	78 %
Shawn Long	3.4	Roll & cut Big	59 %
Myles Turner	3.4	Roll & cut Big	39 %
Montrezl Harrell	3.3	Roll & cut Big	90 %
Stephen Curry	10.9	Movement Shooter	25 %
Gordon Hayward	8.3	Movement Shooter	35 %
J. J. Redick	7.5	Movement Shooter	65 %
Bradley Beal	6.9	Movement Shooter	57 %
Marco Belinelli	6.8	Movement Shooter	86 %
Nikola Jokic	8.6	Post-up Big	99 %
Blake Griffin	7.8	Post-up Big	44 %
Karl-Anthony Towns	7.1	Post-up Big	90 %
Greg Monroe	7.1	Post-up Big	90 %
Kevin Love	6.5	Post-up Big	64 %

Table 11: (continued)

Player	Offensive plus-minus	Position	Position probability
Andre Iguodala	4.9	Spot-up Shooter	68 %
Kyle Anderson	4.8	Spot-up Shooter	43 %
Derrick Jones Jr.	4.7	Spot-up Shooter	23 %
Rondae Hollis-Jefferson	4.7	Spot-up Shooter	26 %
Juancho Hernangomez	4.5	Spot-up Shooter	81 %
Player	Defensive plus-minus	Position	
Rudy Gobert	4.7	Big	
Hassan Whiteside	4.3	Big	
Nene	4.3	Big	
Anthony Davis	4.2	Big	
DeAndre Jordan	4.2	Big	
P. J. Tucker	4.1	Forward	
Paul Millsap	3.8	Forward	
Draymond Green	3.7	Forward	
Marcus Morris Sr.	3.7	Forward	
Jonathon Simmons	3.5	Forward	
Andre Iguodala	0.9	Guard	
Chris Paul	0.9	Guard	
Jrue Holiday	0.8	Guard	
George Hill	0.7	Guard	
Andre Roberson	0.7	Guard	

Table 12: NBA 2017–18 top five plus-minus ratings by position.

Player	Offensive plus-minus	Position	Position probability
Bobby Portis	5.6	Skilled Big	48 %
Channing Frye	5	Skilled Big	71 %
Nikola Mirotic	4.9	Skilled Big	45 %
Frank Kaminsky	4.9	Skilled Big	65 %
DeMarcus Cousins	4.6	Skilled Big	27 %
Chris Paul	7.5	Versatile Scorer	60 %
Damian Lillard	7	Versatile Scorer	76 %
Russell Westbrook	6.7	Versatile Scorer	75 %
James Harden	6.6	Versatile Scorer	48 %
Austin Rivers	6.2	Versatile Scorer	75 %
Eric Gordon	6	Secondary Attacker	39 %
Khriston Middleton	5	Secondary Attacker	37 %
Robert Covington	4.2	Secondary Attacker	38 %
Tobias Harris	4.1	Secondary Attacker	38 %
Otto Porter Jr.	4.1	Secondary Attacker	47 %
Kemba Walker	6.6	Pick & roll Attacker	81 %
Eric Bledsoe	5.2	Pick & roll Attacker	64 %
Fred Van Vleet	5.1	Pick & roll Attacker	84 %
Victor Oladipo	5	Pick & roll Attacker	46 %
Goran Dragic	5	Pick & roll Attacker	69 %
Jordan Bell	6	Roll & cut Big	64 %
John Henson	5.2	Roll & cut Big	88 %
Steven Adams	4.5	Roll & cut Big	93 %
Cody Zeller	3.6	Roll & cut Big	77 %
Keon Looney	3.1	Roll & cut Big	87 %

Table 12: (continued)

Player	Offensive plus-minus	Position	Position probability
Stephen Curry	6.7	Movement Shooter	47 %
Allen Crabbe	4.9	Movement Shooter	90 %
Klay Thompson	4.8	Movement Shooter	52 %
Kyle Korver	4.7	Movement Shooter	58 %
Kentavious Caldwell-Pope	4.6	Movement Shooter	71 %
Karl-Anthony Towns	8.8	Post-up Big	99 %
Nikola Jokic	6.3	Post-up Big	99 %
Al Horford	5.7	Post-up Big	86 %
Anthony Davis	5.4	Post-up Big	68 %
LaMarcus Aldridge	4.3	Post-up Big	70 %
P. J. Tucker	4.5	Spot-up Shooter	67 %
Pascal Siakam	3.7	Spot-up Shooter	66 %
Luke Babbitt	3.6	Spot-up Shooter	34 %
Alex Abrines	3.5	Spot-up Shooter	60 %
Josh Hart	3.5	Spot-up Shooter	48 %
Player	Defensive plus-minus	Position	
Rudy Gobert	4.5	Big	
Anthony Davis	4.3	Big	
Myles Turner	4.3	Big	
Joel Embiid	4.3	Big	
Clint Capela	4.2	Big	
Robert Covington	2.4	Forward	
Jayson Tatum	2.3	Forward	
Davis Bertans	2.2	Forward	
Ersan Ilyasova	2.2	Forward	
David West	2.2	Forward	
Jrue Holiday	5.1	Guard	
Dejounte Murray	4.6	Guard	
Donovan Mitchell	4.5	Guard	
Yogi Ferrell	4.4	Guard	
Kyle Korver	4.3	Guard	

Table 13: NBA 2018–19 top five plus-minus ratings by position.

Player	Offensive plus-minus	Position	Position probability
Kevin Love	6.1	Skilled Big	27 %
Jeff Green	5.9	Skilled Big	82 %
Brook Lopez	5.9	Skilled Big	45 %
Nemanja Bjelica	5.4	Skilled Big	41 %
Alex Len	4.7	Skilled Big	35 %
Kevin Durant	10.1	Versatile Scorer	66 %
James Harden	8.2	Versatile Scorer	32 %
Paul George	7.8	Versatile Scorer	68 %
Damian Lillard	7.4	Versatile Scorer	55 %
Russell Westbrook	7.2	Versatile Scorer	62 %

Table 13: (continued)

Player	Offensive plus-minus	Position	Position probability
Eric Gordon	7.4	Secondary Attacker	86 %
Gordon Hayward	7.4	Secondary Attacker	47 %
Bradley Beal	7.2	Secondary Attacker	26 %
Kyle Kuzma	7.2	Secondary Attacker	37 %
Malcolm Brogdon	6.8	Secondary Attacker	83 %
Mike Conley	7.2	Pick & roll Attacker	47 %
Patty Mills	6.7	Pick & roll Attacker	56 %
Jamal Murray	6.5	Pick & roll Attacker	30 %
Lou Williams	6.4	Pick & roll Attacker	65 %
Trae Young	6.1	Pick & roll Attacker	69 %
John Collins	5.6	Roll & cut Big	44 %
Kevon Looney	4.7	Roll & cut Big	76 %
Dwight Powell	4.7	Roll & cut Big	60 %
Cody Zeller	4.5	Roll & cut Big	91 %
Tristan Thompson	4.5	Roll & cut Big	70 %
Stephen Curry	7.1	Movement Shooter	65 %
Bojan Bogdanovic	6.1	Movement Shooter	73 %
Langston Galloway	5.8	Movement Shooter	56 %
Darius Miller	5.3	Movement Shooter	97 %
Wesley Matthews	5.3	Movement Shooter	73 %
Joel Embiid	8.5	Post-up Big	79 %
Paul Millsap	8.3	Post-up Big	44 %
Andre Drummond	7	Post-up Big	44 %
Al Horford	6.4	Post-up Big	64 %
Marc Gasol	6	Post-up Big	76 %
Danny Green	7.5	Spot-up Shooter	37 %
Danuel House Jr.	7.5	Spot-up Shooter	74 %
Davis Bertans	6.4	Spot-up Shooter	67 %
Luol Deng	6.2	Spot-up Shooter	53 %
Jae Crowder	5.6	Spot-up Shooter	72 %
Player	Defensive plus-minus	Position	
Steven Adams	4.3	Big	
Rudy Gobert	4.3	Big	
Ed Davis	4.3	Big	
Jusuf Nurkic	4.2	Big	
Willie Cauley-Stein	4.2	Big	
Paul George	4.1	Forward	
LeBron James	3.9	Forward	
Royce O'Neale	3.9	Forward	
Maxi Kleber	3.7	Forward	
Giannis Antetokounmpo	3.7	Forward	
Jrue Holiday	1.9	Guard	
Danny Green	1.6	Guard	
Josh Hart	1.3	Guard	
De'Aaron Fox	1.3	Guard	
Kyle Lowry	1.3	Guard	

Table 14: NBA 2020–21 top player plus-minus ratings by position.

Player	Offensive plus-minus	Position	Position probability
Marcus Morris Sr.	5.5	Skilled Big	38 %
Draymond Green	5.2	Skilled Big	41 %
Danilo Gallinari	4.9	Skilled Big	42 %
Paul Reed	4.5	Skilled Big	41 %
P. J. Washington	4.4	Skilled Big	91 %
Kyrie Irving	9.6	Versatile Scorer	94 %
C. J. McCollum	9.4	Versatile Scorer	68 %
Jrue Holiday	9.2	Versatile Scorer	83 %
Luka Dončić	9	Versatile Scorer	77 %
Kawhi Leonard	8.7	Versatile Scorer	79 %
Kyle Lowry	5.8	Secondary Attacker	44 %
Norman Powell	5.7	Secondary Attacker	58 %
Dillon Brooks	5.6	Secondary Attacker	59 %
Marcus Smart	5.6	Secondary Attacker	47 %
Michael Porter Jr.	5.6	Secondary Attacker	41 %
Trae Young	9.5	Pick & roll Attacker	70 %
Ja Morant	8.4	Pick & roll Attacker	64 %
Zach LaVine	8.2	Pick & roll Attacker	62 %
Anfernee Simons	7.9	Pick & roll Attacker	58 %
Reggie Jackson	7.4	Pick & roll Attacker	40 %
Enes Freedom	5.5	Roll & cut Big	57 %
Chris Boucher	4.9	Roll & cut Big	41 %
Kevon Looney	4.5	Roll & cut Big	76 %
Willy Hernangomez	4.5	Roll & cut Big	84 %
Bruce Brown	4.4	Roll & cut Big	29 %
Stephen Curry	8.4	Movement Shooter	71 %
Bradley Beal	7.9	Movement Shooter	30 %
Joe Harris	7.2	Movement Shooter	93 %
Buddy Hield	6.2	Movement Shooter	72 %
Frank Jackson	6.1	Movement Shooter	98 %
Joel Embiid	7.2	Post-up Big	63 %
Zion Williamson	6.7	Post-up Big	24 %
Nikola Jokić	6.5	Post-up Big	94 %
John Collins	5.9	Post-up Big	48 %
Karl-Anthony Towns	5.4	Post-up Big	66 %
Edmond Sumner	5.4	Spot-up Shooter	37 %
Terance Mann	5.1	Spot-up Shooter	44 %
Danny Green	5.1	Spot-up Shooter	55 %
Juancho Hernangomez	4.9	Spot-up Shooter	64 %
Mikal Bridges	4.9	Spot-up Shooter	74 %
Player	Defensive plus-minus	Position	
Rudy Gobert	2.8	Big	
Mike Muscala	2.7	Big	
DeMarcus Cousins	2.6	Big	
Robin Lopez	2.6	Big	
Jusuf Nurkić	2.6	Big	
Paul George	2.5	Forward	
Tobias Harris	2.4	Forward	
Giannis Antetokounmpo	2.4	Forward	
Jimmy Butler	2.3	Forward	
Kawhi Leonard	2.3	Forward	

Table 14: (continued)

Player	Defensive plus-minus	Position
P. J. Dozier	1.4	Guard
Mike Conley	1.1	Guard
Garrett Temple	1	Guard
Michael Carter-Williams	1	Guard
De Andre'Bembry	0.9	Guard

Table 15: NBA 2021–22 top player plus-minus ratings by position.

Player	Offensive plus-minus	Position	Position probability
Aaron Gordon	5.2	Skilled Big	31 %
Karl-Anthony Towns	5.1	Skilled Big	36 %
Danilo Gallinari	4.7	Skilled Big	39 %
Anthony Gill	4.5	Skilled Big	62 %
Jaren Jackson Jr.	4.2	Skilled Big	44 %
Jrue Holiday	11.7	Versatile Scorer	94 %
Jayson Tatum	9.7	Versatile Scorer	86 %
Trae Young	8.3	Versatile Scorer	73 %
Shai Gilgeous-Alexander	8.2	Versatile Scorer	63 %
Darius Garland	8.2	Versatile Scorer	60 %
Gary Trent Jr.	5.7	Secondary Attacker	64 %
Dillon Brooks	5.3	Secondary Attacker	37 %
Miles Bridges	5.1	Secondary Attacker	51 %
Bogdan Bogdanovic	5	Secondary Attacker	47 %
Malik Monk	4.9	Secondary Attacker	84 %
Tyler Herro	7	Pick & roll Attacker	54 %
Cade Cunningham	6.9	Pick & roll Attacker	67 %
LaMelo Ball	6.3	Pick & roll Attacker	92 %
D'Angelo Russell	6.3	Pick & roll Attacker	87 %
Donovan Mitchell	6.2	Pick & roll Attacker	81 %
Jakob Poeltl	5.5	Roll & cut Big	90 %
Clint Capela	5.4	Roll & cut Big	94 %
Isaiah Hartenstein	5.3	Roll & cut Big	97 %
Goga Bitadze	5.2	Roll & cut Big	82 %
Rudy Gobert	5.2	Roll & cut Big	99 %
Stephen Curry	4.9	Movement Shooter	69 %
Jordan Poole	4.8	Movement Shooter	55 %
Garrison Mathews	4.8	Movement Shooter	63 %
Buddy Hield	4.7	Movement Shooter	90 %
Max Strus	4.6	Movement Shooter	94 %
Nikola Jokic	6	Post-up Big	95 %
Joel Embiid	6	Post-up Big	56 %
Bam Adebayo	4.5	Post-up Big	60 %
Jusuf Nurkic	4.5	Post-up Big	90 %
Montrezl Harrell	4.4	Post-up Big	78 %

Table 15: (continued)

Player	Offensive plus-minus	Position	Position probability
Mikal Bridges	4.7	Spot-up Shooter	74 %
Dorian Finney-Smith	4.5	Spot-up Shooter	81 %
Coby White	4.5	Spot-up Shooter	39 %
Lauri Markkanen	4.5	Spot-up Shooter	57 %
Isaac Okoro	4.3	Spot-up Shooter	64 %
Player	Defensive plus-minus	Position	
Jusuf Nurkic	3.4	Big	
Domantas Sabonis	3.3	Big	
Bam Adebayo	3.3	Big	
Anthony Davis	3.2	Big	
Rudy Gobert	3.2	Big	
Paul George	2.6	Forward	
Maxi Kleber	2.5	Forward	
Jayson Tatum	2.5	Forward	
Pascal Siakam	2.4	Forward	
Cameron Johnson	2.4	Forward	
Kenrich Williams	5.1	Guard	
George Hill	4.6	Guard	
Andre Iguodala	3.9	Guard	
Fred Van Vleet	3.7	Guard	
Derrick White	3.7	Guard	

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